

1. Introduction

The "Log Analysis" module has been a pivotal learning experience in understanding the significance of log data in cybersecurity. Logs, as raw data generated by systems, applications, and devices, serve as crucial records for tracking events and activities within an organization's IT environment. This module focused on equipping me with the knowledge and skills necessary to effectively analyze and interpret log data to identify potential security incidents and system anomalies. Through exploring best practices in log analysis, learning about the essential tools available, and gaining hands-on experience with different log management technologies, I have developed a deeper understanding of how log analysis can be used not only for reactive responses but also for proactive threat detection. The course covered everything from parsing log data to leveraging key tools, such as SIEM solutions, to detect, investigate, and mitigate security risks.

2. Module Sections

The screenshot displays the TryHackMe web application interface. At the top, a browser window shows the URL `https://tryhackme.com/room/introtologanalysis`. Below the browser, a terminal window titled "Using GREP to search a logfile for an IP address" shows the command `cmnatic@thm grep "54.36.149.64" logfile.txt` and its output `54.36.149.64`. A green notification box on the right says "Woop woop! Your answer is correct".

Below the terminal, the "Answer the questions below" section contains two questions:

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

super timeline ✓ Correct Answer

Which threat intelligence indicator would `5b31f93c09ad1d065c0491b764d04933` and `763f8bdbc98d105a8e82f36157e98bbe` be classified as?

File Hashes ✓ Correct Answer

At the bottom, a list of tasks is visible:

- Task 4 ☐ Detection Engineering
- Task 5 ☐ Automated vs. Manual Analysis
- Task 6 ☐ Log Analysis Tools: Command Line

TryHackMe | Intro to Log x [TryHackMe] Intro to Log x ThreatFox | Share Indicat... x

https://tryhackme.com/room/introtologanalysis 120%

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard Room progress (22%)

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log

✓ Correct Answer

A log entry containing %2E%2F%2E%2Fproc%2Fse1f%2Fenv1ron was identified. What kind of attack might this infer?

Submit

Task 5 Automated vs. Manual Analysis

Task 6 Log Analysis Tools: Command Line

Task 7 Log Analysis Tools: Regular Expressions

Task 8 Log Analysis Tools: CyberChef

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

TryHackMe | Intro to Log x [TryHackMe] Intro to Log x ThreatFox | Share Indicat... x

https://tryhackme.com/room/introtologanalysis 120%

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard Room progress (22%)

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log

✓ Correct Answer

A log entry containing %2E%2F%2E%2Fproc%2Fse1f%2Fenv1ron was identified. What kind of attack might this infer?

path traversal

✓ Correct Answer

Task 5 Automated vs. Manual Analysis

Task 6 Log Analysis Tools: Command Line

Task 7 Log Analysis Tools: Regular Expressions

Task 8 Log Analysis Tools: CyberChef

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

Room progress (27%)

Reduces the risk of overfitting or false positives on alerts from automated tools.	Events or alerts can be missed! Especially if there
Allows for contextual analysis. The analyst has a broader understanding of the organization and cyber security landscape.	N/A

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

automated

✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Submit

Task 6 ☐ Log Analysis Tools: Command Line

Task 7 ☐ Log Analysis Tools: Regular Expressions

Task 8 ☐ Log Analysis Tools: CyberChef

Room progress (36%)

Reduces the risk of overfitting or false positives on alerts from automated tools.	Events or alerts can be missed! Especially if there
Allows for contextual analysis. The analyst has a broader understanding of the organization and cyber security landscape.	N/A

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated

✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

manual

✓ Correct Answer

Task 6 ☐ Log Analysis Tools: Command Line

Task 7 ☐ Log Analysis Tools: Regular Expressions

Task 8 ☐ Log Analysis Tools: CyberChef

```
File Actions Edit View Help
kali@kali:~/Downloads
$ ls
apache-1691435735822.log  felix.maina095.ovpn
kali@kali:~/Downloads
$ cut -d'"' -f2 apache-1691435735822.log | cut -d"'" -f2 | sort -n
/
/about.php
/admin.php
/contact.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php
kali@kali:~/Downloads
```

TryHackMe | Intro to Log Analysis

Room progress (40%)

solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, and straightforward log analysis tasks.

Woop woop! Your answer is correct

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6

Correct Answer

Hint

In the `apache.log` file, how many total HTTP 200 responses were logged?

--

Submit

Hint

In the `apache.log` file, which IP address generated the most traffic?

Submit

Hint

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

---/---/---:---:--- ---

Submit

Hint

```
kali@kali: ~/Downloads
File Actions Edit View Help
(kali@kali)~/Downloads
$ ls
apache-1691435735822.log  felix.maina@95.ovpn
(kali@kali)~/Downloads
$ cut -d' ' -f2 apache-1691435735822.log | cut -d' ' -f2 | sort -u
/about.php
/admin.php
/contact.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php
(kali@kali)~/Downloads
$ grep '200' apache-1691435735822.log | wc -l
52
(kali@kali)~/Downloads
$
```

TryHackMe | Intro to Log | ThreatFox | Share Indicator | +

https://tryhackme.com/room/introtologanalysis

Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB | OffSec | Academy HTB | TryHackMe | Dashboard

Room progress (45%)

solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, it is more complex to set up and maintain, and may not be necessary for simple and straightforward log analysis tasks.

Woop woop! Your answer is correct

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 ✓ Correct Answer Hint

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 ✓ Correct Answer Hint

In the `apache.log` file, which IP address generated the most traffic?

Submit Hint

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

Submit Hint

```
kali@kali:~/Downloads
ls
apache-1691435735822.log  felix.maina@95.ovpn

(kali@kali)~/Downloads
$ cut -d' ' -f2 apache-1691435735822.log | cut -d' ' -f2 | sort -u

/about.php
/admin.php
/contact.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php

(kali@kali)~/Downloads
$ grep '200' apache-1691435735822.log | wc -l
52

(kali@kali)~/Downloads
$ awk '{print $1}' apache-1691435735822.log | sort | uniq -c | sort -nr | head -1
8 145.76.33.201

(kali@kali)~/Downloads
$
```

TryHackMe | Intro to Log | ThreatFox | Share Indicator | +

https://tryhackme.com/room/introtologanalysis

Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB | OffSec | Academy HTB | TryHackMe | Dashboard

Room progress (50%)

solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, t and straightforward log analysis tasks.

Woop woop! Your answer is correct

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 ✓ Correct Answer Hint

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 ✓ Correct Answer Hint

In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201 ✓ Correct Answer Hint

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

__/__/____:__:__ ____ Submit Hint

Task 7 Log Analysis Tools: Regular Expressions

```
kali@kali: ~/Downloads
ls
apache-1691435735822.log  felix.maina@95.ovpn

(kali@kali)~/Downloads
$ cut -d'***' -f2 apache-1691435735822.log | cut -d' ' -f2 | sort -u

/about.php
/admin.php
/contact.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php

(kali@kali)~/Downloads
$ grep '200' apache-1691435735822.log | wc -l
52

(kali@kali)~/Downloads
$ awk '{print $1}' apache-1691435735822.log | sort | uniq -c | sort -nr | head -1
8 145.76.33.201

(kali@kali)~/Downloads
$ grep '110.122.65.76' apache-1691435735822.log | grep '/login.php'
110.122.65.76 - - [31/Jul/2023:12:34:40 +0000] "GET /login.php HTTP/1.1" 200 9876 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/95.0.4638.110 Safari/537.36"
```

TryHackMe | Intro to Log x [TryHackMe] Intro to Log x ThreatFox | Share Index: x +

https://tryhackme.com/room/introtologanalysis

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard

Room progress (54%)

solution, like the Elastic (ELK) Stack or Splunk, can be more efficient and offer additional log analysis and visualization features. However, a more straightforward log analysis tasks.

Woop woop! Your answer is correct

Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

c701d43cc5a3acb9b5b04db7f1be94f6 ✓ Correct Answer Hint

In the `apache.log` file, how many total HTTP 200 responses were logged?

52 ✓ Correct Answer Hint

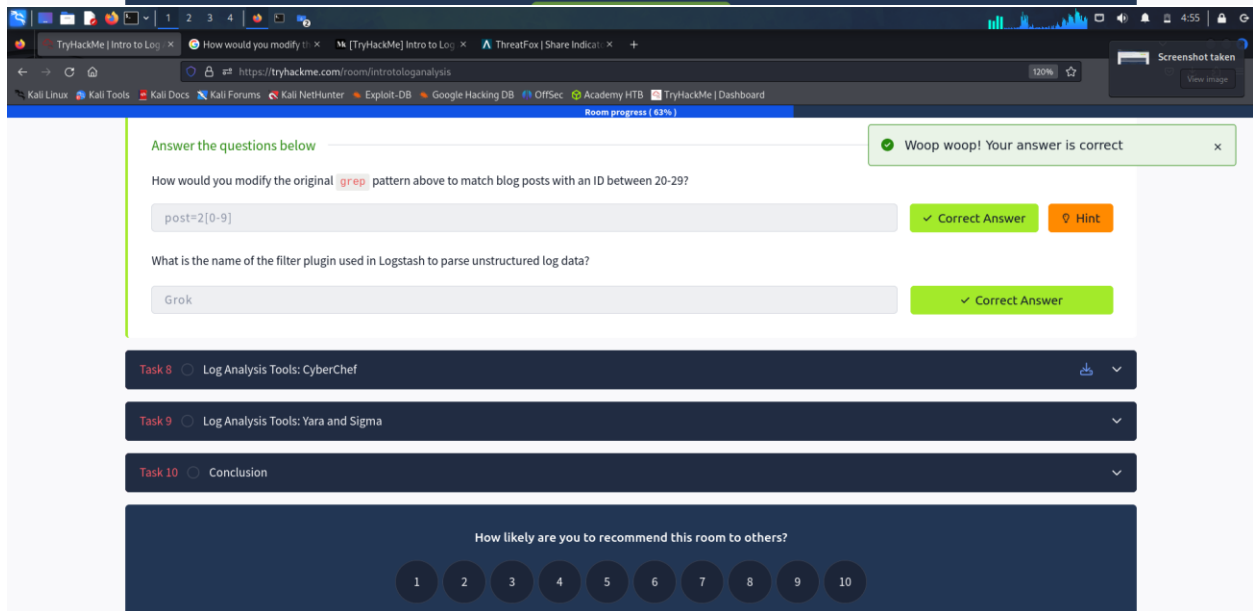
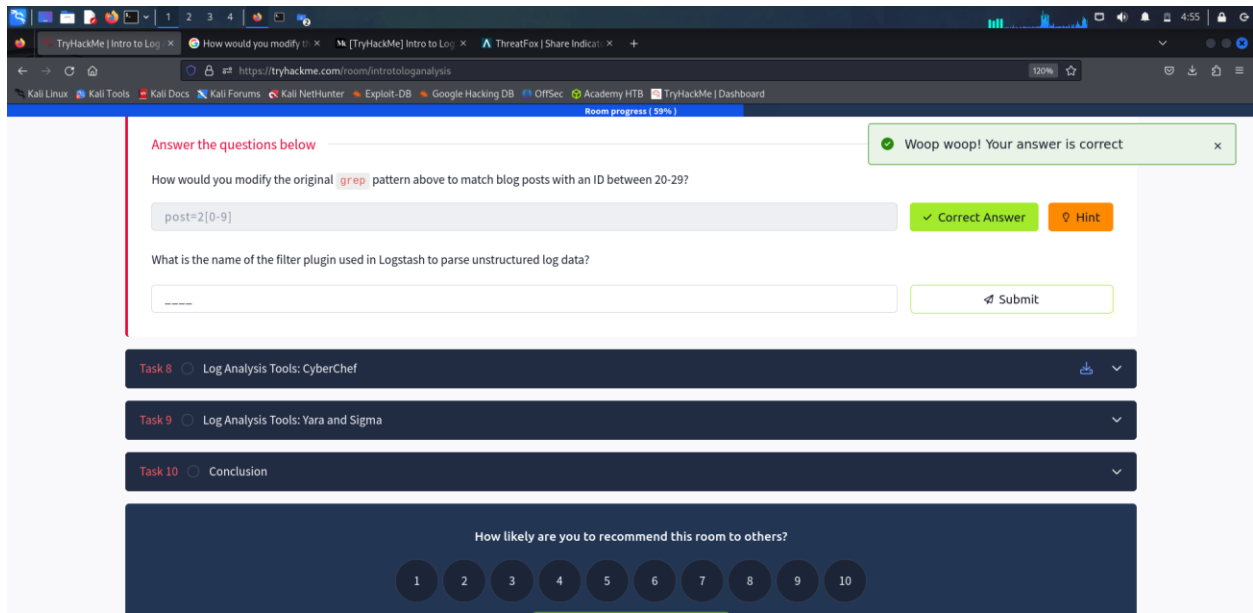
In the `apache.log` file, which IP address generated the most traffic?

145.76.33.201 ✓ Correct Answer Hint

What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

31/Jul/2023:12:34:40 +0000 ✓ Correct Answer Hint

Task 7 Log Analysis Tools: Regular Expressions



Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

✔ Woop woop! Your answer is correct x

Complete

✓ Correct Answer

Submit

Submit

Task 10 ○ Conclusion

TryHackMe | Intro to Log x Regular expression - Cyb... x How would you modify it x [TryHackMe] Intro to Log x ThreatFox | Share Indica... x

https://gchq.github.io/CyberChef/#recipe=Regular_expression('Strings',(GET)/*%3D,true,false,false,false,'list matches')&input=MTQzLjExMC4yMjY2IC0gLSBhMjcvU2VwL...

Download CyberChef Last build: 2 days ago - Version 10 is here! Read about the new features here Options About / Support

Operations

- rege
- Regular expression
- Register
- To Case Insensitive Regex
- From Case Insensitive Regex
- Render Image
- Resize Image
- Rotate Image
- Pseudo-Random Number Generator
- From MessagePack
- Parse IP range
- Normalise Image
- Parse User Agent
- Sharpen Image
- Generate Image
- JMServer Fingerprint
- HASSH Server Fingerprint

Recipe

Regular expression

Built in regexes

Regex (GET) /. * =

Strings

☒ Case insensitive ☒ ^ and \$ match at newlines ☐ Dot matches all

☐ Unicode support ☐ Astral support ☐ Display total

Output format: List matches

Input

File details

Name: access.log
Size: 19,051 bytes
Type: unknown
Loaded: 100%

Output

GET /dns-query?name=dnssec.shadowserver.org&type=...
GET /XDEBUG_SESSION_START=...
GET /VEHneGNZqkV5Q0hFR19XSvpBUkR9==...
GET /vimeo.sogou?op=B&query=...

STEP **BAKE!** Auto Bake

TryHackMe | Intro to Log x From Base64 - CyberChef x How would you modify it x [TryHackMe] Intro to Log x ThreatFox | Share Indica... x

https://gchq.github.io/CyberChef/#recipe=From_Base64('A-Za-z0-9%2B/%3D',true,false)&input=VKVoTmUwTipRaIZUTBoRUh0VhTvBCVWSOT09

Download CyberChef Last build: 2 days ago - Version 10 is here! Read about the new features here Options About / Support

Operations

- Search...
- Favourites
- To Base64
- From Base64
- To Hex
- From Hex
- To Hexdump
- From Hexdump
- URL Decode
- Regular expression
- Entropy
- Fork
- Magic
- Data format
- Encryption / Encoding
- Public Key
- Arithmetic / Logic

Recipe

From Base64

Alphabet: A-Za-z0-9+/=

☒ Remove non-alphabet chars

☐ Strict mode

Input

VEHneGNZqkV5Q0hFR19XSvpBUkR9==

Output

[THM]{CYBERCHEF_WIZARD}

STEP **BAKE!** Auto Bake

TryHackMe | Intro to Log | From Base64 - CyberChef | How would you modify it | [TryHackMe] Intro to Log | ThreatFox | Share Indicator | +

Room progress (77%)

Answer the questions below

Locate the "loganalysis.zip" file under `/root/.Rooms/introLoganalysis/task8` and extract the contents.

No answer needed

✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145

✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM[CYBERCHEF_WIZARD]

✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

Submit

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

TryHackMe | Intro to Log | From Base64 - Regular expression | How would you modify it | [TryHackMe] Intro to Log | ThreatFox | Share Indicator | +

https://github.com/CyberChef/RecipeFrom_Base64(A-Za-z0-9%2B%3D,true,false)Regular_expression(MAC address:[A-Fa-f\d]{2}[?::-][A-Fa-f\d]{2}[S]?true,true,false,false)

Download CyberChef

Last build: 2 days ago - Version 10 is here! Read about the new features here

Options About / Support

Operations

Search...

Favourites

To Base64

From Base64

To Hex

From Hex

To Hexdump

From Hexdump

URL Decode

Regular expression

Entropy

Fork

Magic

Data format

Encryption / Encoding

Public Key

Arithmetic / Logic

Recipe

From Base64

Alphabet A-Za-z0-9+/=

Remove non-alphabet chars

Strict mode

Regular expression

Built in regexes

Regex [A-Fa-f\d]{2}[?::-][A-Fa-f\d]{2}[S]?

MAC address

Case insensitive

^ and \$ match at newlines

Dot matches all

Unicode support

Astral support

Display total

Output format List matches

Input

Output

38:2E:9A:4B:74:01

STEP

BAKE!

Auto Bake

TryHackMe | Intro to Log | x From Base64, Regular e... x How would you modify i... x [TryHackMe] Intro to Log x ThreatFox | Share Indica... x +

https://tryhackme.com/room/introtologanalysis 120%

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard

Room progress (81%)

Answer the questions below

Locate the "loganalysis.zip" file under `/root/.Rooms/introtologanalysis/task8` and extract the contents.

No answer needed ✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145 ✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM[CYBERCHEF_WIZARD] ✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61 ✓ Correct Answer

Task 9 Log Analysis Tools: Yara and Sigma

Task 10 Conclusion

TryHackMe | Intro to Log | x From Base64, Regular e... x How would you modify i... x [TryHackMe] Intro to Log x ThreatFox | Share Indica... x +

https://tryhackme.com/room/introtologanalysis 120%

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec Academy HTB TryHackMe | Dashboard

Room progress (86%)

- IP Addresses based on a range (for example, an ASN or a subnet)
- IP addresses in HEX
- If an IP address lists more than a certain amount (i.e., alert if an IP address is found five times)
- And combined with other rules. For example, if an IP address visits a specific page or does a certain action

If you want to learn more about Yara, check out the [Yara](#) room on TryHackMe.

Answer the questions below

What languages does Sigma use?

YAML ✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

Submit

What keyword is used to denote the "name" of a rule in YARA?

Submit

Task 10 Conclusion

TryHackMe | Intro to Log | From Base64, Regular e... | How would you modify i... | [TryHackMe] Intro to Log | ThreatFox | Share Indica... | +

https://tryhackme.com/room/introtologanalysis

Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB | OffSec | Academy HTB | TryHackMe | Dashboard

Room progress (90%)

- IP Addresses based on a range (for example, an ASN or a subnet)
- IP addresses in HEX
- If an IP address lists more than a certain amount (i.e., alert if an IP address is found five times)
- And combined with other rules. For example, if an IP address visits a specific page or does a certain action

If you want to learn more about Yara, check out the [Yara](#) room on TryHackMe.

Answer the questions below

What languages does Sigma use?

YAML

✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title

✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

Submit

Task 10 Conclusion

TryHackMe | Intro to Log | From Base64, Regular e... | How would you modify i... | [TryHackMe] Intro to Log | ThreatFox | Share Indica... | +

https://tryhackme.com/room/introtologanalysis

Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB | OffSec | Academy HTB | TryHackMe | Dashboard

Room progress (95%)

- IP Addresses based on a range (for example, an ASN or a subnet)
- IP addresses in HEX
- If an IP address lists more than a certain amount (i.e., alert if an IP address is found five times)
- And combined with other rules. For example, if an IP address visits a specific page or does a certain action

If you want to learn more about Yara, check out the [Yara](#) room on TryHackMe.

Answer the questions below

What languages does Sigma use?

YAML

✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title

✓ Correct Answer

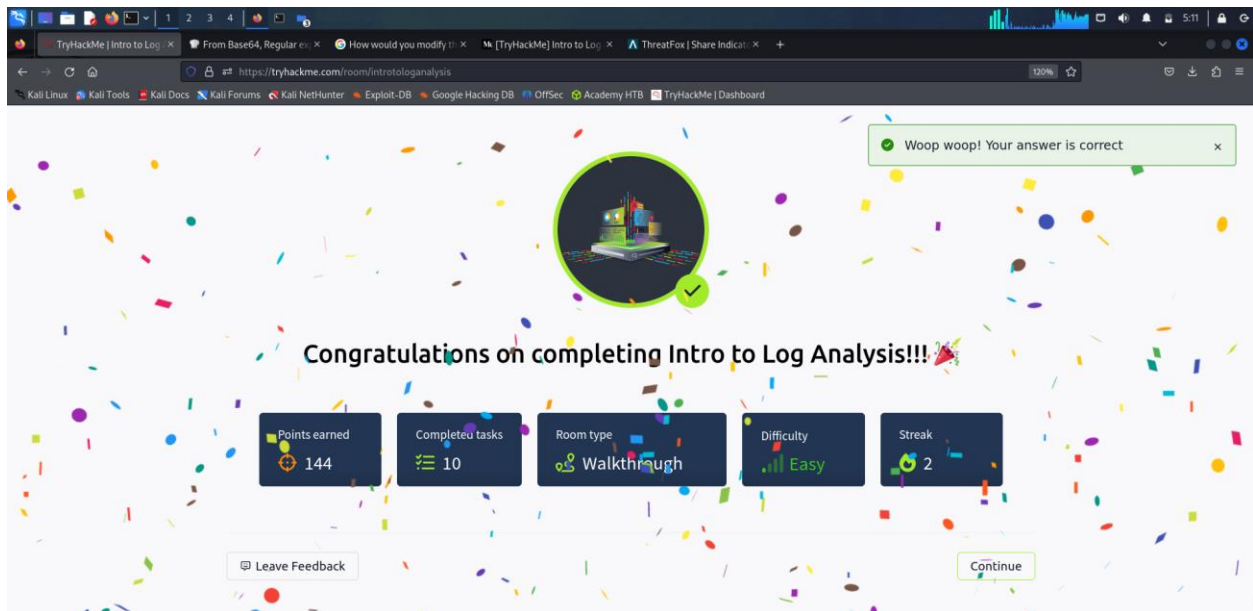
What keyword is used to denote the "name" of a rule in YARA?

rule

✓ Correct Answer

Task 10 Conclusion

3. Shareable Link and Final Screenshot



4. Social Media Posting

https://x.com/intent/post?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fintrotologanalysis%3Futm_source%3Dtwitter%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&text=Share%20your%20achievement&via=realtryhackme&hashtags=tryhackme

https://www.facebook.com/sharer/sharer.php?u=https%3A%2F%2Ftryhackme.com%2Froom%2Fintrotologanalysis%3Futm_source%3Dfacebook%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&t=Share%20your%20achievement

https://www.linkedin.com/feed/?shareActive=true&text=Share%20your%20achievement%0D%0Ahttps%3A%2F%2Ftryhackme.com%2Froom%2Fintrotologanalysis%3Futm_source%3Dlinkedin%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom

5. Conclusion

Upon reflection, the "Log Analysis" module has significantly enhanced my ability to approach cybersecurity with a data-driven mindset. I now fully appreciate how crucial effective log management and analysis are in maintaining the security posture of an organization. The best practices for log collection, retention, and analysis have provided me with a structured approach to handling large volumes of log data, ensuring that important events are not overlooked. Furthermore, using tools to automate log parsing and event correlation has given me hands-on experience in streamlining the detection process, which is critical for identifying security threats before they escalate.

What I've learned extends beyond technical expertise—this module has honed my analytical thinking. I've gained an understanding of how to interpret raw log data within the broader context of a system's behavior and security infrastructure. The ability to identify patterns, recognize anomalies, and correlate

events from multiple sources will undoubtedly be valuable in both real-time and retrospective investigations.

Despite this progress, I recognize that mastering log analysis requires continual practice and familiarization with new tools and evolving attack vectors. I plan to further develop my skills by diving deeper into log automation techniques, exploring advanced SIEM configurations, and understanding the specific logging protocols for different environments. The knowledge I have gained so far has solidified my interest in log analysis and security operations, and I look forward to continuing to build on this foundation in my cybersecurity journey.